Causal Discovery with Latent Confounders on PetShop Microservice Latency: ABIC vs SPOT

Project - Milestone 2: Technical Foundation

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The Problem: Who Causes Whose Latency?

- Microservice apps are chains of services calling each other
- A slow database makes every service above it look slow too
- Goal: learn the causal structure from monitoring data alone
- Why it matters: root-cause diagnosis needs the origin of a slowdown, not just the symptoms
- Difficulty: some causes are never measured (unmonitored components, shared load)

Dataset: PetShop (Hardt et al., Amazon)

Scenarios: low_traffic, high_traffic, temporal_traffic1, temporal_traffic21

Representation: Acyclic Directed Mixed Graph (ADMG)

Nodes: observed service latencies

$A \rightarrow B$ A's latency directly affects B's latency

$A \leftrightarrow B$ A and B share an unobserved common cause

Model: linear-Gaussian, two weight matrices

D = directed edge weights

B = bidirected edge (noise covariance) weights

$\Sigma = (I - \delta)^{-T} \cdot \beta \cdot (I - \delta)^{-1}$

A plain DAG would turn a hidden cause into a wrong directed edge. The ADMG represents it with $\leftrightarrow$ instead.

PetShop Data and Ground Truth

- Variable: Average latency per service, per scenario

  Observed variables: d = [d] Samples: n = [n]

- Ground truth from the call graph, inverted:

  callee → caller (a callee's latency propagates upward)

- Services with no usable metric are treated as latent
- Ground-truth pipeline: DAG over observed + latent

  → latent projection → ADMG → MAG → PAG (Tetrad)

- Preprocessing: z-scoring by default, Gaussian-rank transform available, raw as sensitivity check

Learning(1): ABIC (Baseline)

Parameter learning

- RICF: maximum-likelihood weights for a fixed graph
- Used to compute BIC

Structure learning

- Maximize likelihood subject to "graph is an ancestral ADMG"
- Constraint made differentiable: $h(D, B) = \mathrm{tr}(e^D) - d + \mathrm{sum}(e^D \circ B)$, $h = 0 \Leftrightarrow$ valid graph
- Augmented Lagrangian + L-BFGS-B
- Restarts chosen by BIC, then threshold weights to get edges

Our implementation: hand-derived analytic gradients, checked against finite differences (error below 1e-8)

State Of The Art: SPOT (Microsoft Research)

Ma, Ding, Fu, Zhang, Wang, Han, Zhang.
"Scalable Differentiable Causal Discovery in the Presence of Latent Confounders with Skeleton Posterior." KDD 2024. Microsoft Research and HKUST.

Problem: ABIC is slow and inaccurate beyond ~20 nodes
Idea: estimate a posterior over the skeleton (which pairs are adjacent) and use it to guide ABIC's optimization

Paper results (synthetic, 50-100 nodes, 10 datasets): Skeleton F1 Arrowhead F1 Tail F1
ABIC 0.84 0.76 0.67 SPOT 0.91 0.86 0.78

Paper evaluates on synthetic data and Sachs protein data.
It does not test microservice data.

Learning(2): Skeleton Posterior (Stage 2b)

Goal: p_ij = P(i and j are adjacent | data), for every pair

Method: amortized inference

- Simulate random graphs + linear-Gaussian data with known skeleton
- Features: Fisher-z conditional-independence statistics, Markov-blanket features, percentile summaries
- Model: cascade of 3 gradient-boosting classifiers (orders 0, 1, 2)

Held-out result (60 unseen simulated graphs, d = 25-40)

AUROC 0.989 | AUPRC 0.946 | Log-loss 0.057 | Brier 0.015 | ECE 0.013

Runtime (single core)

Offline training: 203 s (once)

Online inference: 0.29 s per dataset

Caveat: evaluated on simulated data only. PetShop is out of distribution for this model.

A Discrepancy We Found in The Released Code

Discrepancy: Selection condition inverted in released feature code

A small p-value means "dependent", but the released feature code keeps pairs when **pval > α**, which selects independent pairs instead.

Metric	Correct (p ≤ α)	As released (p > α)
Order-0 screen keeps true edges	**98.1%**	1.9%
Markov-blanket recall of neighbours	**65.9%**	1.1%
End-to-end AUROC	**0.988**	0.975

Scope: effect in OUR re-implementation. Effect on the paper's own numbers is unknown.

SPOT: One Optimizer, One Difference

SPOT = ABIC + a gradient mask (Stage 2c)

ABIC and SPOT call the same optimizer. The only difference is the mask.

Update rule (paper Eq. 9), at every L-BFGS evaluation:

- Weakening update (moving toward zero): always kept
- Strengthening update: kept with probability (p_ij + c)^(1/(t+1)), c = 0.1, t = ALM outer iteration Rejected coordinates get gradient 0

Hard floor (in released scripts, not in paper text): pairs with p ≤ floor are pinned to 0

- As released: 1e-3
- Calibrated: 6e-3 (chosen once, on separate held-out graphs)

Sanity check: with p = 1 and no floor, SPOT equals ABIC exactly

Experimental design and KPIs

Baseline: ABIC
Proposed: SPOT (released floor and calibrated floor)
SOTA reference: SPOT paper

Ablations: random-pair posterior, oracle posterior, mask only, floor only

Structure KPIs: skeleton, arrowhead, tail F1, TPR, FDR against the PAG; corrected and legacy versions

Efficiency KPIs: optimizer time, L-BFGS evaluations, convergence Runtime in three parts: offline training, online inference, optimization

Protocol: paired comparison, same data and seeds, mean ± SD, Wilcoxon signed-rank test
Hardware: single core for synthetic timings

Synthetic results

Initial results: synthetic data (d = 25, n = 1000, 1 restart, 6 graphs, corrected metrics)

Data	Method	Skel F1	Head F1	Tail F1	Time (s)
Raw	ABIC	0.894	0.806	0.704	75.7
Raw	SPOT (as released)	0.949	0.878	0.751	55.3
Raw	**SPOT-cal**	**0.961**	**0.916**	**0.892**	**24.0**
Z-scored	ABIC	0.687	0.390	0.104	82.9
Z-scored	SPOT (as released)	0.718	0.387	0.117	96.1
Z-scored	**SPOT-cal**	**0.730**	**0.419**	**0.106**	**59.4**

Raw: SPOT-cal beats ABIC on skeleton F1 on 6 of 6 graphs

Z-scored: 5 of 6 graphs, +0.042 mean, not significant (p = 0.094)

Bidirected edges: Both methods find ~23-27 bidirected edges on z-scored data (true graphs have ~3)

Ablations: where does the gain come from?

Z-scored synthetic data (3 graphs)

Variant	Skel F1	Head F1	Tail F1	Time (s)
ABIC	0.635	0.308	0.065	83.6
SPOT (as released)	0.722	0.379	0.107	105.1
SPOT-cal	**0.688**	**0.340**	**0.132**	**72.8**
Mask only (no floor)	0.671	0.345	0.064	81.9
Floor only (no mask)	0.667	0.321	0.104	73.0
Random posterior	0.330	0.149	0.088	37.6
Oracle posterior	0.922	0.586	0.206	20.7

Posterior information: Random posterior is far worse than ABIC, oracle is far better. The learned posterior recovers only part of the oracle's headroom.

Efficiency drivers: Speed-up comes mostly from the floor, not the stochastic mask.

Graph constraints: With only 3 graphs, mask-only vs floor-only performance cannot be fully separated.

Working pipeline: demo

1. python scripts/gradcheck.py analytic vs finite-difference gradients
2. python tests/run_tests.py automated tests (mask rule, SPOT = ABIC at p = 1, ...)
3. results/syn_2c_summary.md aggregated tables from raw per-run records
4. Load models/skeleton_posterior.pkl post.predict(X) gives p_ij in 0.29 s
5. One PetShop scenario, end to end [after 2d]

PetShop Run Setup (2d)

Benchmark Dataset Configurations

SCENARIO	n	d	HIDDEN NODES	TRUE EDGES (DIR / BIDIR)	AUTOCORR (LAG-1 MEDIAN)
low_traffic	589	29	13	30 / 1	0.26
high_traffic	589	35	9	38 / 0	0.29
temporal_traffic1	1652	31	10	32 / 1	0.33

Preprocessing & Optimization Parameters

- **Preprocessing:** Gaussian-rank transform + z-score.
- **Algorithm Setup:** Same data, thresholds, 1 restart, and L-BFGS settings applied for ABIC and SPOT.
- **Seeds Configuration:** 3 / 3 / 2 (varying optimizer only; single dataset per scenario).
- **Posterior Training Range:** Samples n ∈ [300, 2000], autocorrelation up to 0.6.
- **Validation Coverage:** All three active scenarios strictly fit within the target range.

Execution & Scenario Pipeline Status

COMPLETED — **3 of 4 scenarios** processed (main comparison complete).

PENDING
- **temporal_traffic2** scenario
- **PetShop ablations**
- **Equal-budget** testing
- **Raw-data sensitivity** analysis

PetShop Headline Results (2d)

PetShop Results: ABIC vs SPOT (Corrected Metrics, Mean Over Seeds: 3 / 3 / 2)

SCENARIO	METHOD	SKEL F1	HEAD F1	TAIL F1
high_traffic	ABIC	0.513	0.299	0.167
	SPOT (as released)	0.536	0.258	0.123
	SPOT-cal	**0.537**	**0.301**	**0.181**
low_traffic	ABIC	0.484	0.288	0.000
	SPOT (as released)	0.514	0.298	0.000
	SPOT-cal	**0.521**	**0.243**	**0.000**
temporal_traffic1	ABIC	0.544	0.317	0.000
	SPOT (as released)	0.554	0.334	0.067
	SPOT-cal	**0.590**	**0.407**	**0.133**

Across Scenarios (Scenario Means)

METHOD	SKEL F1	HEAD F1	TAIL F1
ABIC	0.514	0.301	0.056
SPOT	0.535	0.297	0.063
SPOT-cal	**0.550**	**0.317**	**0.105**

Key Takeaways & Findings

- **SPOT-cal skeleton F1 higher** in 3 of 3 scenarios (**+0.036 mean**).
- **Seed-to-seed SD:** about 0.02–0.03 (optimizer only); no significance claim.

PetShop Skeleton Posterior Transfer (2d)

Posterior Transfer Performance Across Scenarios

METRIC	SYNTHETIC (HELD-OUT)	LOW_TRAFFIC	HIGH_TRAFFIC	TEMPORAL1
Posterior AUROC	0.99	0.83	0.93	0.88
Posterior AUPRC	0.95	0.52	0.59	0.57
True edges pinned by floor	~1%	32%	16%	15%

Estimated vs True Graph (Per Run Averages)

SCENARIO	DIRECTED (EST/TRUE)	BIDIRECTED (EST/TRUE)
`low_traffic`	~10-15 / 30	~20-25 / 1
`high_traffic`	~12-14 / 38	~32-39 / 0
`temporal1`	~7-9 / 32	~28-34 / 1

Key Takeaways & Ablation Status

- **Clear Domain Shift:** Skeleton posterior is noticeably weaker on PetShop datasets.
- **Recall Cap:** The floor mechanism rules out true edges, directly capping recall.
- **Edge Substitution:** Both methods substitute bidirected edges in place of directed ones.

PetShop Ablations:

- Random / oracle, mask-only / floor-only: **not yet run**.
- Synthetic ablations are available on **slide 12**.

Corrected vs Legacy Metrics (2e)

Corrected vs Legacy Metrics (PetShop, 3 Scenarios, Scenario Means)

METHOD	SKELETON F1 CORR | LEGACY	ARROWHEAD F1 CORR | LEGACY	TAIL F1 CORR | LEGACY
ABIC	0.514 | 0.514	0.301 | 0.479	0.056 | 0.271
SPOT	0.535 | 0.535	0.297 | 0.471	0.063 | 0.350
SPOT-cal	**0.550 | 0.550**	**0.317 | 0.488**	**0.105 | 0.385**

Key Takeaways & Findings

- **Skeleton F1** is identical in both versions.
- **Legacy metrics** inflate arrowhead F1 by about **0.17** and tail F1 by **0.2–0.3**.
- **Method ranking** is unchanged: **SPOT-cal** leads ahead of ABIC on skeleton, with a small edge on arrowheads.
- **SPOT (released floor)** is slightly below ABIC on arrowheads in both versions.

Runtime breakdown (2e)

Runtime on PetShop (single core, 1 restart, all runs converged)

METHOD	OFFLINE TRAINING	ONLINE INFERENCE	OPTIMIZATION (S) [LOW / HIGH / TEMPORAL1 / MEAN]
ABIC	none	none	83 / 192 / 47 / 115
SPOT (released)	203 s (once)	~0.3-0.4 s	32 / 92 / 46 / 58
SPOT-cal	**203 s (once)**	**~0.3-0.4 s**	**19 / 78 / 39 / 46**

Key Highlights & Efficiency

- **Optimization Speedup:** SPOT-cal optimization time is **0.41x ABIC** on high_traffic (mean 46 s vs 115 s).
- **Break-even Threshold:** Reached in **~3 datasets** including offline training (~69 s saved per dataset).
- **Convergence:** All runs converged successfully under the single-core setup.

Caveats & Experimental Context

- **Speed-up Source:** Speed-up comes mostly from floor pruning pairs, not the mask.
- **Single-Core Constraint:** Not directly comparable to paper's 32 cores (paper: ABIC 3139 s, SPOT 2883 s).
- **Pending Benchmarks:** Equal wall-clock budget runs are not yet run.

Interpretation of PetShop results (2e)

What happened
SPOT-cal beats ABIC on skeleton F1 in 3 of 3 scenarios (+0.036), within optimizer-seed spread. No reliable arrowhead or tail gain. About 2-2.5x faster optimization. Both methods score ~0.5 skeleton F1.

Expected?
Partly. Matches our synthetic z-scored result; much smaller than the paper's synthetic gain.

Why
- Posterior is out of distribution: AUROC 0.83-0.93 vs 0.989
- Calibrated floor pins 15-32% of true edges (vs ~1% simulated)
- Both methods return 17-41 bidirected edges vs 0-1 true, so arrowhead and tail scores mostly measure mislabelled directed edges
- Sample size and autocorrelation are inside the training range, so skewed, non-Gaussian, call-graph structure is the likelier cause (not yet isolated)

Limitations

- PetShop: 3 scenarios, 2-3 optimizer seeds, 1 restart; seeds do not vary the data, so no significance claims
- Not run: temporal_traffic2, PetShop ablations, equal-budget, raw-data check, FCI/RFCI baseline
- Posterior out of distribution on PetShop (AUROC 0.83-0.93); the calibrated floor, tuned on simulations, is too aggressive there
- Violated assumptions: non-Gaussian margins, time series treated as i.i.d.; feedback from retries cannot be represented
- Bidirected over-use: 17-41 estimated vs 0-1 true
- Part of SPOT's speed-up comes from a hard floor not in the paper text
- Inference layer: unit-tested only; accuracy on real incidents untested
- Ground truth is the inverted call graph, not guaranteed to equal the statistical dependence structure of the data
- Synthetic evidence is small (6 graphs; 3 for ablations)

Next Steps

Next steps

Milestone 3:
- Root-cause evaluation on real PetShop issue runs
  (top-1, top-3 recall, MRR; anomaly vs dag vs admg;
  ABIC, SPOT, truth graphs)
- Complete remaining PetShop runs: seeds 3-4, ablations,
  equal-budget, raw vs z-scored; confirm temporal_traffic2
- FCI/RFCI baseline
- Statistical tests; error analysis of missed and spurious edges
- Posterior suited to PetShop-like data (or domain adaptation)

Milestone 4:
- Final tables, robustness, reproducibility package,
  final SOTA positioning
- Empirical support for the ADMG justification

Conclusions and contributions :

Conclusions

- Working foundation: latent-aware PetShop ground truth, ABIC and SPOT on one verified optimizer, skeleton posterior, corrected metrics, root-cause inference layer
- Posterior: synthetic held-out AUROC 0.989; found and measured an inverted test in the released code
- Synthetic: SPOT clearly beats ABIC on raw data (skeleton F1 0.961 vs 0.894); small and not significant on z-scored data
- PetShop (3 scenarios): SPOT-cal skeleton F1 0.550 vs ABIC 0.514, higher in 3 of 3, within optimizer-seed spread; no reliable arrowhead or tail gain; about 2-2.5x faster optimization
- Main finding: the posterior is out of distribution on PetShop (AUROC 0.83-0.93), which limits SPOT's benefit

Contributions
Anant: SPOT implementation, report
Ankit: benchmarks (PetShop and synthetic runs, ablations, tables)
Milan: inference layer, ADMG justification, powerpoint presentation



Thank You!

Causal Discovery with Latent Confounders on PetShop Microservice Latency: ABIC vs SPOT

Course: CSE516 - Probabilistic Graphical Models